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**Analysis of Selected SLAM Methods
Using Various Sensor Types**

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Abstract in Polish

Abstract in English

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Chapter 1

Introduction

Simultaneous Localisation and Mapping (SLAM) is one of the fundamental problems in mobile robotics [1]. It combines two tasks: localisation which estimates the robot's position and orientation, and mapping which builds a spatial model of the surrounding environment. In SLAM, neither is assumed to be known in advance, so both must be estimated at the same time from noisy sensor data. Without a map the robot cannot plan paths or recognise places it has visited before and without knowing its own position it cannot build a correct map. Since GPS is unavailable in most indoor environments, SLAM is essential for any robot that must work autonomously indoors.

The difficulty of SLAM problem comes from mutual dependency of two subtasks. A correct map can only be built if the robot knows its position, and position can only be estimated if there is a map to match against. Even with a good algorithm, small sensor errors add up over time. The key correction mechanism is loop closure [2]: when the robot returns to a place it has already visited, it recognises the match and uses it to correct the drift that has built up in its trajectory. Without it, errors grow without bound and SLAM algorithms fail.

The problem was first formalised in 1986 by Smith, Self, and Cheeseman [3] and independently by Durrant-Whyte [4], who described how to reason probabilistically about robot pose and map uncertainty. The term SLAM was coined at the International Symposium on Robotics Research (ISRR) in 1995 [5]. Early solutions used Extended Kalman Filters, Rao-Blackwellised Particle Filters, and maximum likelihood estimation. Research then focused on observability, convergence, and consistency, which led to the first widely used open-source libraries [2]. Supported sensors have since expanded from 2D laser scanners to 3D LiDARs, cameras, and tightly-coupled inertial systems. The current focus has shifted to robust long-term

operation.

SLAM has been evolving since its formulation four decades ago, driven largely by the growing use of mobile robots in indoor environments such as homes or warehouses, where GPS is not available. Today it is used across many domains: robotic vacuum cleaners navigate home environments, autonomous vehicles map urban roads, unmanned aerial vehicles use it for inspection and surveillance, and underwater drones can monitor coral reefs or detect mines.

Despite four decades of progress, SLAM is not a fully solved problem [2]. It can work reliably for slow-moving robots in static indoor environments, but can struggle in fast-moving or highly dynamic outdoor scenarios. Many open-source implementations exist, each making different design choices with different trade-offs. Choosing the right algorithm for a given robot and environment is not obvious, which motivates a direct, controlled comparison.

1.1 Goal and Scope

The goal of this thesis is to evaluate eleven SLAM algorithms under identical conditions on a simulated, differential drive robotic platform equipped with a 3D LiDAR, a stereo camera, and an IMU. The algorithms cover the main families: sensor fusion, LiDAR odometry, 2D and 3D LiDAR SLAM, tightly-coupled LiDAR-IMU SLAM, and visual SLAM.

All algorithms are tested on a single dataset recorded in NVIDIA Isaac Sim, which provides exact ground-truth poses. All experiments run under ROS 2 (Robot Operating System).

Trajectory accuracy is measured using Absolute Trajectory Error (ATE) and Relative Pose Error (RPE) against the simulator ground truth. Map quality is assessed with Chamfer distance and F-score for algorithms that produce a 3D point cloud, and visually for those that produce an occupancy grid. Computational cost is measured as mean CPU load and peak RAM usage during execution.

1.2 Thesis Structure

Chapter 2 contains the theoretical background required to understand the thesis: sensors, map representations, point cloud registration, state estimation, graph-based SLAM, loop closure detection, and evaluation metrics.

Chapter 3 describes each of the eleven evaluated algorithms, grouped by method family.

Chapter 4 presents the experimental setup, the dataset, and the results for all eleven algorithms, including trajectory accuracy, map quality, and computational cost.

Chapter 5 concludes the thesis and provides a summary of findings made during experimental evaluation of algorithms.

Chapter 2

Theoretical Background

2.1 Navigation in Mobile Robotics

Navigation is "the ability of the robot to act based on its knowledge and sensor values so as to reach its goal positions as efficiently and as reliably as possible" [6, p. 257]. It can be decomposed into five components:

1. Perception: recording and interpreting raw sensor data such as point clouds or images.
2. Localisation: estimating the robot pose within a known environment.
3. Mapping: constructing a spatial model of the environment from sensor observations.
4. Path planning: computing a collision-free route from the current position to a goal.
5. Motion control: generating actuator commands that follow the planned path while reacting to unexpected obstacles.

SLAM combines components 2 and 3, estimating the robot pose and the map simultaneously without prior knowledge. This thesis focuses on SLAM, therefore path planning and motion control are not evaluated. The robot trajectory is pre-recorded and replayed from a dataset.

2.1.1 Mapping

Mapping is the problem of building a spatial model of the environment from sensor measurements, assuming the robot trajectory is known [7, p. 221]. Even with

exact poses the problem is non-trivial. Sensor noise and visually similar places can cause incorrect data association, and any pose error introduces geometrical distortion across the map. Common map representations used in this thesis are the 2D occupancy grid and the 3D point cloud, described in Section 2.3.

2.1.2 Localisation

Localisation is the complementary problem to mapping. Given a known map, the robot determines its current pose by fusing sensor measurements with motion commands [7, p. 157]. Because pose cannot be read from a sensor directly, the estimate is built incrementally as each new measurement updates the belief. All algorithms evaluated in this thesis operate with a known starting position (position tracking mode) [7, p. 159]: the search is confined to a small neighbourhood of the current estimate, which keeps computational cost low.

2.1.3 SLAM

SLAM assumes that neither the robot poses $\mathbf{x}_{1:t}$ nor the map $\mathbf{m}$ are known, and estimates both jointly from sensor measurements $\mathbf{z}_{1:t}$ and motion commands $\mathbf{u}_{1:t}$ [7, p. 245]. Two forms of the problem are common [7, pp. 245 and 246]. Online SLAM discards past poses and maintains only the current pose together with the map:

$$p(\mathbf{x}_t, \mathbf{m} \mid \mathbf{z}_{1:t}, \mathbf{u}_{1:t}). \quad (2.1)$$

Full SLAM keeps all past poses and estimates the complete trajectory together with the map:

$$p(\mathbf{x}_{1:t}, \mathbf{m} \mid \mathbf{z}_{1:t}, \mathbf{u}_{1:t}). \quad (2.2)$$

The online form is recovered from the full one by integrating out past poses. The two dominant solution families of SLAM problem, filter-based and graph-based methods, are described in Sections 2.5 and 2.6 respectively.

2.2 Sensors

Sensors provide the data necessary for SLAM implementation. This thesis uses four sensor types: a 3D LiDAR, a stereo camera, an IMU, and wheel odometry encoders.

2.2.1 LiDAR

A LiDAR (Light Detection And Ranging) sensor measures distances by emitting laser pulses and recording the round-trip travel time Δt . The distance to the

reflecting surface is given by:

$$d = \frac{c \cdot \Delta t}{2}, \quad (2.3)$$

where c is the speed of light. A rotating 3D LiDAR fires pulses across multiple vertical channels while spinning horizontally. Each full revolution produces a cloud of tens of thousands of (x, y, z) points. Its main advantages for SLAM are accuracy and independence from ambient lighting. The main drawback is continuous movement of the robot during a single scan. As a result, successive points are captured from slightly different poses, making the resulting cloud geometrically inconsistent without correction.

2.2.2 Stereo Camera

A stereo camera consists of two horizontally separated cameras with a known baseline B between them. When the same scene point is visible in both images it appears at slightly different horizontal positions. This horizontal offset d allows the depth Z to be computed as

$$Z = \frac{f \cdot B}{d}, \quad (2.4)$$

where f is the focal length in pixels.

Cameras capture dense appearance information from which features can be extracted for feature matching and place recognition. However, if the environment lacks distinctive texture, stereo vision fails to find reliable correspondences [6, p. 113].

2.2.3 IMU

An Inertial Measurement Unit (IMU) combines a three-axis accelerometer and a three-axis gyroscope. Both outputs are corrupted by a slowly varying bias $\mathbf{b}$ and noise $\boldsymbol{\eta}$:

$$\tilde{\mathbf{a}} = \mathbf{a} + \mathbf{b}_a + \boldsymbol{\eta}_a, \quad \tilde{\boldsymbol{\omega}} = \boldsymbol{\omega} + \mathbf{b}_g + \boldsymbol{\eta}_g, \quad (2.5)$$

where $\tilde{\mathbf{a}}$ and $\tilde{\boldsymbol{\omega}}$ are the measured acceleration and angular velocity, and $\mathbf{a}$, $\boldsymbol{\omega}$ are the true values. The IMU runs at a higher rate than other sensors, providing motion information between slower LiDAR or camera updates. However, integrating corrupted signals accumulates errors over time, so the IMU must be combined with another sensor to remain accurate [7, p. 91].

2.2.4 Wheel Odometry

Wheel encoders measure the angular velocity of each wheel, from which the linear speeds v_L and v_R of the left and right wheels are calculated. For a differential-drive robot moving at heading θ , the pose increment over a time step Δt is given by:

$$\Delta x = \frac{v_L + v_R}{2} \Delta t \cos \theta, \quad \Delta y = \frac{v_L + v_R}{2} \Delta t \sin \theta, \quad \Delta \theta = \frac{v_R - v_L}{L} \Delta t, \quad (2.6)$$

where L is the distance between the wheels. Integrating these increments over time provides a pose estimate. However, wheel slip and calibration errors in the wheel radius and distance between them causes the estimate to drift over time [6, Ch. 3].

2.2.5 360-Degree Camera

A 360-degree (omnidirectional) camera captures the full horizontal field of view in a single frame, either through a fisheye lens or by stitching multiple camera feeds into one projection. The full environment coverage reduces the risk of losing tracked features during sharp rotations. The main drawbacks are strong lens distortion and the absence of a single agreed sensor model, since the term covers fisheye, catadioptric (camera with mirror), and multi-camera designs without a unified definition. This sensor type is not evaluated in this thesis: stereo camera-based visual SLAM is used instead, and Isaac Sim does not provide a native omnidirectional camera [6, Ch. 4].

2.3 Map Representations

The map representation affects memory usage, update speed, and available downstream operations. The algorithms evaluated in this thesis produce either a 2D occupancy grid or a 3D point cloud.

2.3.1 Occupancy Grid

An occupancy grid divides the 2D plane into a regular grid of square cells, each storing the probability $p(m_i \mid \mathbf{z}_{1:t}, \mathbf{x}_{1:t})$ that cell m_i is occupied, given all measurements and poses up to time t [7]. Rather than storing the probability directly, cells are updated using the log-odds representation

$$l_{t,i} = \log \frac{p(m_i \mid \mathbf{z}_{1:t}, \mathbf{x}_{1:t})}{1 - p(m_i \mid \mathbf{z}_{1:t}, \mathbf{x}_{1:t})}, \quad (2.7)$$

Cells with log-odds above a threshold are marked occupied (obstacles), those below a second threshold are marked free, and the rest remain unknown. Occupancy grids directly support path planning, as each cell indicates whether the region is available for the robot. The representation is limited to 2 dimensions and the cell size, fixed as a parameter before mapping, determines the spatial resolution of the map.

2.3.2 3D Point Cloud

A 3D point cloud is an unordered set of points in a common reference frame, each defined by its (x, y, z) coordinates [7, p. 221]. A global map is built by registering each incoming LiDAR scan into the map frame using the estimated robot pose and accumulating the resulting points. The resulting map captures the full three-dimensional geometry of the environment.

Point clouds are well suited for geometric inspection and for evaluating map quality against a reference scan. Their drawbacks are high memory consumption in large environments and the absence of information about free-space: unlike an occupancy grid, a point cloud records only observed surfaces, not which regions are reachable.

Visual SLAM methods do not accumulate dense range measurements. Instead, they build sparse point clouds by triangulating matched image features into 3D landmarks. The density of such a map depends on the visual texture of the environment, therefore it is unsuitable for measurable geometric evaluation and is not evaluated as such in this thesis.

2.4 Point Cloud Registration

Point cloud registration finds the rigid transformation that best aligns two overlapping point clouds. In LiDAR SLAM each new scan is aligned to the previous one or to the accumulated map to estimate motion.

2.4.1 K-d Tree

Many registration and evaluation methods need to find the closest point in a target cloud for each point in a source cloud. Comparing against every stored point is too slow when clouds contain tens of thousands of points.

A k-d tree [9] solves this by organising points into a hierarchical structure that allows most of the search space to be skipped, reducing the cost from $O(n)$ to $O(\log n)$ per query. ICP and Chamfer Distance both rely on this structure.

2.4.2 Iterative Closest Point (ICP)

ICP [10] is a general method for registering 3D shapes. The algorithm alternates between two steps. First, each point in the source cloud is matched to its nearest neighbour in the target cloud. Second, the rigid transformation that minimises the sum of squared distances between matched pairs is computed. These steps repeat until the change in error falls below a predefined threshold. The solution can get trapped in a local minimum rather than finding the globally correct alignment. A poor initial estimate leads to wrong point pairs being formed, causing the algorithm to converge to an incorrect result.

2.4.3 Point-to-Plane ICP

Point-to-plane ICP [11] extends standard ICP by replacing the point-to-point distance with the distance to the local surface at each matched target point. The surface is estimated from the neighbouring points of the target using a surface normal. This means only motion away from the surface is penalised, while motion along it is unconstrained. In environments with many walls, floors, and ceilings this leads to faster convergence than standard ICP. MOLA Odometry uses point-to-plane ICP.

2.4.4 Generalized ICP (GICP)

GICP [12] extends point-to-plane ICP by treating both the source and the target as locally flat surfaces rather than just the target. Each point carries a small description of the surface patch around it, and matching minimises the misalignment between these patches on both sides. This makes the algorithm less sensitive to noisy matches and to the threshold used for accepting a correspondence. GICP is the default registration method of KISS-ICP.

2.4.5 Normal Distributions Transform (NDT)

NDT [13] avoids finding point pairs entirely. The target cloud is divided into a 3D voxel grid. For each voxel, the average position and the spread of all points inside it are computed, describing where the surface is and how flat or scattered it is. To find the alignment, each source point is transformed and the algorithm checks how well it fits into the surface description of the voxel it lands in. The transformation is adjusted until the source points fit the target descriptions as well as possible. Because the fit measure changes smoothly as the transformation changes, the optimisation is more stable than ICP, which can jump when matched pairs suddenly switch. The fixed voxel grid also makes memory usage predictable

and the surface descriptions average out noise from uneven point density. MOLA Odometry uses NDT.

2.5 Filter-Based State Estimation

Filter-based methods maintain a probability distribution over the robot state and update it recursively as new sensor data arrives. The update cycle alternates between a prediction step, which propagates the state forward using a motion model, and a correction step, which incorporates the latest sensor measurement to reduce uncertainty.

2.5.1 Extended Kalman Filter (EKF)

The standard Kalman filter assumes that both the motion model and the sensor model are linear, which is rarely true in robotics. The EKF [7, pp. 48 to 65] solves this by approximating the nonlinear models locally at each step, making it applicable to real robot motion and sensor fusion. It maintains a best estimate of the robot state together with a measure of confidence in that estimate. The filter alternates between two steps. In the prediction step, the motion model is used to estimate where the robot moved to, causing the confidence to decrease. In the correction step, a new sensor measurement is used to pull the estimate toward what the sensor observed, and the confidence increases. In this thesis the EKF is used by the `robot_localization` package to fuse wheel odometry and IMU data into a dead-reckoning pose estimate.

The iterated variant (iEKF) [14] improves on this by repeating the correction step several times, each time re-approximating the sensor model at the latest corrected estimate rather than the original one. This gives a more accurate result when the sensor model is strongly nonlinear. FAST-LIO2 uses an iEKF to correct pose, velocity, and IMU biases simultaneously.

2.5.2 Rao-Blackwellized Particle Filter

The robot state in a Rao-Blackwellized Particle Filter [15] is represented as a set of particles, each containing a possible trajectory and a separate map built from that trajectory. When the robot moves, all particles are updated according to the motion model, each in a slightly different way to reflect motion uncertainty. When a new sensor scan arrives, each particle checks how well the scan fits its own map and receives a score. Particles with a poor score are discarded and replaced by copies of high-scoring ones, so the set concentrates on the most plausible trajectories. The

final map is taken from the best-scoring particle. GMapping, evaluated in this thesis, applies this approach to occupancy-grid SLAM [32].

2.6 Graph-Based SLAM

Most SLAM systems are structured as a two-stage pipeline. The front end processes raw sensor data to extract relative pose constraints (through scan matching, visual feature tracking, or IMU integration) and identifies loop closure candidates. The back end takes these constraints as input and optimises the full trajectory and map for global consistency, typically by solving a nonlinear least-squares problem over a pose or factor graph. In contrast to filter-based methods, which process measurements sequentially and discard past information, graph-based approaches retain the full trajectory and can enforce global consistency whenever a loop closure is detected [16].

2.6.1 Pose Graphs

A pose graph [16] represents the robot trajectory as a set of nodes, one for each robot pose, connected by edges. Each edge records the relative movement between two poses as measured by odometry or scan matching, together with a confidence value. When a loop closure is detected, a new edge is added connecting the current pose to the earlier matching pose. The graph is then optimised to find the set of poses that best satisfies all constraints simultaneously, correcting the drift accumulated along the trajectory. Cartographer 2D, SLAM Toolbox, and RTABMap all use pose graph back-ends.

2.6.2 Keyframes

A keyframe is a selected frame from the robot's trajectory that is stored as a reference point for matching and optimisation. Not every frame becomes a keyframe. A new keyframe is typically created only when the robot has moved or rotated enough since the last one. Storing only keyframes keeps the pose graph small and the optimisation fast, without losing the information needed to correct drift. In the pose graph, each node usually corresponds to a keyframe. IMU measurements between two keyframes are accumulated into a single summary, and visual odometry uses a small group of recent keyframes to refine the current pose estimate.

2.7 Loop Closure Detection

Loop closure detection [2] identifies when the robot has returned to a previously visited location. Small errors accumulate with each step, so the estimated position can drift significantly over a long trajectory. When a revisit is detected, a new edge is added to the pose graph connecting the current pose to the earlier one, and the graph is re-optimised to distribute the correction across the full trajectory. How candidates are identified depends on the algorithm. SLAM Toolbox searches nearby nodes in the pose graph. Cartographer searches all previously built submaps using branch-and-bound, described below. RTABMap compares a compact descriptor of the current scan against all stored nodes, so it can detect a revisit even after a long detour. For visual systems, bag-of-words place recognition is used, described in Section 2.7.2.

2.7.1 Branch-and-Bound Search

Cartographer uses branch-and-bound [18] to find the best alignment between an incoming scan and a stored submap. The pose space is divided into blocks. For each block, the algorithm estimates the best score that any pose inside it could achieve. If that estimate is already worse than the current best, the entire block is discarded without further search. This guarantees a globally optimal result while avoiding an exhaustive search, unlike ICP which depends on the starting position and can miss the best answer.

2.7.2 Bag-of-Words Visual Place Recognition

Bag-of-words (BoW) place recognition [19] represents each image as a compact list of visual patterns and how often they appear. These patterns, called visual words, are learned in advance from a large set of images. When the robot sees a new place, its image is converted into this representation and compared against all stored images in the map. This comparison is fast because it involves only counting matching visual words rather than comparing raw pixel values. A stored image that scores high enough becomes a loop closure candidate. Before accepting it, the system verifies that the two images can be geometrically aligned. Only if that check passes is the loop closure edge added to the pose graph. RTABMap uses this approach for loop closure detection.

2.8 Visual SLAM Fundamentals

Visual SLAM estimates the robot trajectory from image sequences by detecting and tracking sparse visual features rather than measuring range directly. This thesis evaluates feature-based stereo systems, which recover depth from stereo disparity (Section 2.2) and estimate pose from the 3D positions of matched keypoints.

2.8.1 Stereo Vision and Triangulation

Stereo depth is computed from the disparity between the two images as given by Equation (2.4) in Section 2.2. Each matched feature pair is converted to a 3D landmark by tracing both camera rays and finding their intersection [21, Ch. 11]. Depth uncertainty grows quadratically with distance, so visual SLAM front-ends typically discard landmarks beyond a sensor-specific threshold.

2.8.2 Feature Detection and Description

Visual feature detection finds distinctive points in an image and describes each one in a way that allows it to be recognised in later frames.

A good feature to track is a corner, not an edge. On an edge, a feature can slide along it without the tracker noticing. A corner is uniquely located and stays put. The Good Features to Track (GFTT) detector [22] selects points where the image intensity changes strongly in all directions, not just one. GFTT is commonly used in visual SLAM systems where reliable tracking is more important than distinctive appearance description.

ORB (Oriented FAST and Rotated BRIEF) [23] is a fast alternative to heavier descriptors such as SIFT and SURF. It finds corners, determines the local patch orientation, and builds a compact binary fingerprint using that orientation to achieve rotation invariance. Matching is done by counting differing bits between two fingerprints, which is fast to compute. Running the detector at multiple zoom levels gives scale robustness. ORB is widely used in visual SLAM systems.

2.8.3 Visual Odometry

Visual odometry estimates the robot's motion by tracking features across consecutive camera frames [24]. Matched features are converted to 3D landmarks and used to compute the pose change between frames. A small group of recent keyframes is kept and jointly refined to reduce accumulated error. Without loop closure, small errors add up with every step, so drift grows steadily over a long trajectory.

2.9 Evaluation Metrics

Two groups of metrics are used in this thesis. Trajectory metrics measure how closely the estimated robot path matches the ground truth poses recorded from the simulation. Map metrics measure the geometric accuracy of the 3D point cloud produced by each algorithm relative to the reference map from the simulator. The ground truth is a time-stamped sequence of $SE(3)$ poses, one per sensor frame, taken directly from the simulator with no sensor noise.

Each metric produces a per-pose or per-point error value. These are summarised by three statistics. The RMSE (Root Mean Square Error) is the square root of the average squared error and is the primary reported value, as it penalises large errors more than small ones. The mean is the simple average of all error values. The maximum is the single largest error in the sequence and indicates worst-case behaviour.

2.9.1 Umeyama Alignment

A SLAM algorithm starts its own coordinate frame from the first sensor measurement, so the estimated trajectory may be shifted or rotated relative to the ground truth. Before measuring errors, the two trajectories are aligned using the method of Umeyama [25], which finds the rotation and translation that bring them as close together as possible. For LiDAR-based systems the scale is fixed to one, so only the orientation and position offset are corrected.

2.9.2 Absolute Trajectory Error (ATE)

The Absolute Trajectory Error measures the global geometric consistency of the estimated path [26]. After Umeyama alignment, the translational error at each time step i is the Euclidean distance between the aligned estimated position $\hat{\mathbf{p}}_i$ and the ground truth position $\mathbf{p}_i^*$. The RMSE over all n poses is

$$\text{ATE} = \sqrt{\frac{1}{n} \sum_{i=1}^n \|\hat{\mathbf{p}}_i - \mathbf{p}_i^*\|^2}. \quad (2.8)$$

A large ATE indicates that the algorithm accumulates drift over the full trajectory, or that loop closure is absent or ineffective. In addition to the RMSE, the mean and maximum translational errors are reported to characterise the error distribution.

2.9.3 Relative Pose Error (RPE)

The Relative Pose Error measures local odometric accuracy independently of the global alignment [26]. For each consecutive pose pair $(i, i+1)$ the relative motion is computed both in the estimate and in the ground truth, and the translational part of the residual between the two relative poses is taken as the local error. The RMSE over all such pairs is

$$\text{RPE} = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n-1} \|\mathbf{e}_i^{\text{trans}}\|^2}, \quad (2.9)$$

where $\mathbf{e}_i^{\text{trans}}$ is the translational component of the residual between the relative pose in the estimate and the relative pose in the ground truth over the same one-step window. Because RPE is computed over short, fixed-length windows it captures short-term drift that ATE may hide through loop-closure corrections.

2.9.4 Chamfer Distance

The Chamfer Distance (CD) is a symmetric measure of geometric fidelity between two point clouds [27]. Given an estimated cloud $\mathcal{A}$ and a reference cloud $\mathcal{B}$, it averages the mean nearest-neighbour distance in both directions:

$$\text{CD}(\mathcal{A}, \mathcal{B}) = \frac{1}{2} \left(\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \min_{\mathbf{b} \in \mathcal{B}} \|\mathbf{a} - \mathbf{b}\| + \frac{1}{|\mathcal{B}|} \sum_{\mathbf{b} \in \mathcal{B}} \min_{\mathbf{a} \in \mathcal{A}} \|\mathbf{b} - \mathbf{a}\| \right). \quad (2.10)$$

The first term penalises spurious points in the estimate that have no counterpart in the reference. The second term penalises incomplete regions of the map. CD is reported in metres and reflects the overall geometric accuracy of the reconstructed map, but is sensitive to outlier points, which can inflate the score without reflecting a systematic reconstruction failure.

2.9.5 F-score

The F-score at a distance threshold d combines precision and recall into a single bounded quality measure [28]. Precision measures how accurate the estimated map is: a high precision means most points in the estimate are close to the reference. Recall measures how complete the estimated map is: a high recall means most parts of the reference are covered by the estimate. A point $\mathbf{a} \in \mathcal{A}$ is considered correct if its nearest neighbour in $\mathcal{B}$ is at most d away, and vice versa:

$$P(d) = \frac{1}{|\mathcal{A}|} \left| \{ \mathbf{a} \in \mathcal{A} : \min_{\mathbf{b}} \|\mathbf{a} - \mathbf{b}\| \leq d \} \right|, \quad (2.11)$$

$$R(d) = \frac{1}{|\mathcal{B}|} \left| \{ \mathbf{b} \in \mathcal{B} : \min_{\mathbf{a}} \|\mathbf{b} - \mathbf{a}\| \leq d \} \right|, \quad (2.12)$$

$$F(d) = \frac{2 P(d) R(d)}{P(d) + R(d)}. \quad (2.13)$$

Three thresholds are used in this thesis: 5 cm, 10 cm, and 20 cm. A tight threshold such as 5 cm only counts a point as correctly placed if it is very close to the reference. A loose threshold such as 20 cm is more forgiving and rewards overall completeness. The F-score is always between 0 and 1, where 1 means a perfect match.

Chapter 3

Algorithm description

This chapter describes each of the 10 algorithms evaluated in this thesis, grouped by method family. Table 3.1 lists them with their sensors and method families.

Table 3.1: Evaluated algorithms overview

#	Algorithm	Sensors	Method family
1	EKF Dead Reckoning	Wheel + IMU	Sensor fusion
2	KISS-ICP	LiDAR	LiDAR odometry
3	MOLA Odometry	LiDAR	LiDAR odometry
4	Cartographer 2D	LiDAR (2D scan)	2D LiDAR SLAM
5	GMapping	LiDAR (2D scan)	2D LiDAR SLAM
6	SLAM Toolbox	LiDAR (2D scan)	2D LiDAR SLAM
7	RTABMap (LiDAR)	LiDAR	3D LiDAR SLAM
8	FAST-LIO2	LiDAR + IMU	LiDAR-IMU odometry
9	cuVSLAM	Stereo camera	Visual SLAM
10	RTABMap RGB-D	Stereo camera	Visual SLAM

3.1 Sensor Fusion

Sensor fusion combines wheel odometry and IMU to estimate pose without building a map. It serves as a dead-reckoning baseline.

3.1.1 EKF Dead Reckoning

EKF (Extended Kalman Filter) node from the `robot_localization` ROS2 package fuses wheel odometry and IMU data to estimate the robot's pose without any geometric map. Dead reckoning accumulates drift from wheel slip and noisy IMU data, so errors grow unboundedly over time unless corrected.

3.2 LiDAR Odometry

LiDAR odometry methods estimate the motion of the robot by aligning consecutive LiDAR scans using the Iterative Closest Point (ICP) algorithm. Each new scan is compared to a local map of the previous few scans. The best-fitting transformation is taken as the pose change.

These methods do not build a global map and do not detect loop closures, which leads estimation error to accumulate proportionally to travelled distance.

LiDAR odometry is computationally efficient and works without visual features. Methods that add loop closure or IMU fusion should outperform it especially on longer trajectories.

3.2.1 KISS-ICP

KISS-ICP [29] (Keep It Small and Simple ICP) uses GICP with adaptive parameters, requiring no manual tuning. It was published in 2022 by researchers at the University of Bonn.

3.2.2 MOLA Odometry

MOLA [30] (Modular Optimization framework for Localization and mApping) was developed at the University of Almería. Only its LiDAR odometry component (`mola_lidar_odometry`) was used here.

It uses point-to-plane ICP with an NDT voxel map. Older scans are discarded as the robot moves to limit memory usage. MOLA is built as a modular pipeline, which makes components easy to swap but requires more configuration than KISS-ICP.

3.3 2D LiDAR SLAM

2D LiDAR SLAM is designed for planar laser scanners that produce a single horizontal sweep of range measurements. Representing the environment as a horizontal plane reduces the pose estimation problem from 6-DoF to 3-DoF (x ,

y, θ). This simplification lowers computational cost and is valid for wheeled robots on flat floors, but it discards all height information. 2D SLAM is therefore unsuitable when full spatial reconstruction is needed or when the robot operates on uneven terrain. The TurtleBot3 platform evaluated in this thesis is a wheeled robot operating on a flat floor, so the planar LiDAR is sufficient.

The simulation used in this thesis does not have a 2D laser scanner, only a point cloud from a 3D LiDAR. The lack of 2D LiDAR in simulation is caused by computational limitations. Data from 3D LiDAR was processed offline: one horizontal ring was extracted from each 3D scan and saved into a separate pre-processed bag. All three 2D SLAM algorithms read from this bag.

Cartographer and SLAM Toolbox both build pose graphs and run global optimisation when a loop closure is detected. GMapping uses a particle filter, where each particle carries its own map.

3.3.1 Google Cartographer 2D

Cartographer 2D [18] was developed by Google and uses a two-stage architecture. The local stage matches each incoming scan against a small occupancy submap. Once a submap is complete it is frozen and becomes a node in the pose graph. The global stage searches for loop closures between frozen submaps using a branch-and-bound matcher, which is globally optimal. Accepted loop closures are added as edges and the pose graph is then optimised. The final map is a single occupancy grid assembled from all submaps.

3.3.2 SLAM Toolbox

SLAM Toolbox [31] was developed by Steve Macenski and is the recommended 2D SLAM solution for ROS 2 Navigation2. Its defining feature is lifelong mapping: the pose graph and map can be serialised to disk and resumed in a later session, which makes it well suited to large environments that cannot be mapped in a single run.

Scan matching uses a correlation-based matcher: a grid of candidate poses is evaluated and the one that maximises the overlap score between the incoming scan and the stored map is selected as the best alignment. Unlike Cartographer's branch-and-bound approach, this can get stuck in local minima, making loop closure more sensitive to tuning and environment geometry.

3.3.3 GMapping

GMapping [32] is a Rao-Blackwellized particle filter SLAM. There is no explicit loop closure. When the robot revisits a known area, particle weights naturally shift toward poses consistent with the existing map, which corrects the trajectory. This makes GMapping more tolerant of repetitive environments where a correlation-based matcher might commit to a false loop, but it also limits correction ability on long trajectories.

GMapping officially targets ROS 1. The ROS 2 port used here was compiled from source against the `openslam_gmapping` library.

3.4 3D LiDAR SLAM

3D LiDAR SLAM builds maps from the full point cloud without discarding spatial information, making it suitable when the environment cannot be described by a single horizontal cross-section.

3.4.1 RTABMap (LiDAR)

RTABMap [33] (Real-Time Appearance-Based Mapping) is a graph-based SLAM system developed at Université de Sherbrooke. Pose estimates come from ICP-based scan matching against a local point cloud map. The pose graph is globally optimised whenever a loop closure is detected.

Loop closure detection is appearance-based. The system compares descriptors (compact numerical summaries of the scan's geometric structure) of the current scan against all stored nodes in the map, not just nearby ones. This means a revisit can be detected even after a long detour, which methods that only search nearby nodes would miss.

To keep computation bounded, memory is split into Working Memory (WM) and Long-Term Memory (LTM). When processing time exceeds a set threshold, the least-visited nodes are moved to LTM. They are only retrieved when a loop closure candidate matches them.

3.5 Tightly-Coupled LiDAR-IMU

These algorithms work with full 3D point clouds with an IMU fused directly inside the same estimator.

The LiDAR provides geometric accuracy to estimate and remove the IMU’s accumulated drift. Neither sensor alone is sufficient.

One system is evaluated: FAST-LIO2, an odometry-only system with no loop closure.

A rotating LiDAR produces points over a full revolution while the robot is moving, so points captured at the start and end of the scan come from slightly different poses. Deskewing corrects this by transforming all points in a scan to a common timestamp using IMU-interpolated poses. IMU preintegration [20] avoids re-integrating all high-frequency IMU measurements from scratch at every step by accumulating them into a single compact motion estimate that remains valid during optimisation. FAST-LIO2 relies on both techniques.

3.5.1 FAST-LIO2

FAST-LIO2 [34] was developed at the HKU-MARS laboratory. It registers raw LiDAR points directly to the map. No feature extraction step is needed. This makes it work well with LiDARs that have a small field of view or non-repetitive scanning patterns, where feature-based methods fail.

It uses an iEKF: each point is matched to its nearest neighbours, a local plane is fitted, and the point-to-plane distance drives the update, correcting pose and IMU biases in one step.

The map is stored in an incremental k-d tree (ikd-Tree), which extends the standard k-d tree (Section 2.4.1) to support inserting and deleting points without rebuilding the structure. This keeps search fast as the map grows.

3.6 Visual SLAM

Visual SLAM estimates the trajectory from image sequences using feature-based stereo vision. The fundamental weakness is texture dependency: flat or uniformly coloured surfaces produce few keypoints and the system loses tracking.

Three systems are evaluated, all using a stereo camera pair.

3.6.1 cuVSLAM

cuVSLAM [36] is NVIDIA’s visual SLAM library, distributed as part of the Isaac ROS ecosystem. It runs the entire pipeline on an NVIDIA GPU, which allows dense feature detection and tracking at rates that would be too slow on CPU. Stereo matching produces depth for each tracked feature via triangulation. IMU

measurements are fused with the visual estimates to improve tracking stability and robustness to motion blur. Pose increments are estimated from the tracked 3D feature positions and IMU data, and a pose graph back-end with loop closure maintains global consistency. Because the library is closed-source, internal parameters beyond those exposed through the ROS 2 interface are not accessible. The system requires a compatible NVIDIA GPU to run.

3.6.2 RTABMap RGB-D

RTABMap [33] in RGB-D mode reuses the same graph SLAM back-end as the LiDAR variant but uses a visual front-end for loop closure. In this evaluation the depth image comes from stereo matching on the Hawk camera. Wheel odometry provides the motion prior between frames. Loop closure candidates are identified both by spatial proximity in the pose graph and by bag-of-words matching of visual features extracted from the RGB image. Accepted loop closures trigger a global graph optimisation that corrects the accumulated drift. The accumulated map is a dense 3D point cloud built by projecting depth images into the world frame at each pose.

Chapter 4

Experimental Study

This chapter presents the experimental configuration, the simulation environment, the evaluation methodology, and the results obtained by the eleven evaluated SLAM algorithms. All experiments were conducted on a single dataset to ensure full comparability between algorithms.

4.1 Experimental Setup

4.1.1 Robot Platform

The experimental platform is the TurtleBot3 Burger, a small differential-drive mobile robot designed for indoor navigation. It has a maximum linear velocity of 0.22 m/s and carries its sensors and onboard compute in a compact, lightweight frame. The TurtleBot3 Burger is a widely used research platform and is representative of the class of small ground robots that rely on SLAM for autonomous operation in GPS-denied environments.

4.1.2 Sensor Configuration

The robot is equipped with four sensors that together cover the modalities evaluated in this thesis.

- 3D LiDAR. A rotating 3D LiDAR operating at 33.6 Hz (scan period ≈ 30 ms), producing dense point clouds.
- Stereo camera. A Leopard Imaging LI-AR0234CS-NVIDIA Hawk stereo camera with a 100 mm baseline, 1920×1200 px resolution per lens, and a

121.5° horizontal field of view, recording at 60 fps. A depth map at 640×360 px is also provided.

- IMU. An inertial measurement unit integrated with the camera module, publishing linear acceleration and angular velocity at approximately 62 Hz.
- Wheel odometry. Pose increments derived from motor encoders.

4.1.3 Software Framework

All algorithms are executed under ROS 2 Jazzy on the same hardware. Sensor data are replayed from a recorded rosbag at real-time speed, ensuring identical temporal conditions for every algorithm.

Two preprocessing steps were applied where required by specific algorithms. For the 2D LiDAR SLAM methods (GMapping, Cartographer 2D, and SLAM Toolbox), the 3D point cloud was projected onto a planar 2D scan by retaining only points within a fixed height band around the robot. For LIO-SAM, each point in the cloud was assigned a ring index derived from its elevation angle, as the algorithm requires an organised point cloud with per-point ring information. All other algorithms consumed the raw sensor topics without modification.

4.2 Simulation Environment

4.2.1 Environment Description

4.2.2 Dataset Recording

4.2.3 Ground Truth Extraction

4.3 Evaluation Methodology

4.3.1 Trajectory Accuracy

4.3.2 Map Quality

4.3.3 Computational Cost

4.4 Results

Table 4.1: Trajectory and map quality metrics N/A bag_6. ATE and RPE in metres. Row shading: ATE<0.1 m, 0.1 to 2 m, >2 m.

Algorithm	ATE RMSE	ATE mean	ATE max	RPE RMSE	RPE mean	CD (m)	F@5 cm	F@10 cm	F@20 cm
GMapping	0.015	0.013	0.055	0.002	0.000	2.258	0.005	0.041	0.113
DLIO	0.042	0.035	0.376	0.003	0.002	1.598	0.043	0.099	0.196
Cartographer 2D	0.280	0.153	1.059	0.023	0.003	0.103	0.627	0.794	0.867
KISS-ICP	0.505	0.229	3.098	0.023	0.017	0.570	0.053	0.232	0.449
MOLA Odometry	0.223	0.215	0.331	0.003	0.003	0.210	0.016	0.249	0.597
RTABMap LiDAR	0.004	0.004	0.014	0.002	0.002	0.049	0.598	0.951	0.999
SLAM Toolbox	4.942	4.447	7.430	0.013	0.003	2.990	0.003	0.025	0.133
RTABMap RGB-D	0.102	0.052	0.690	0.007	0.000	N/A	N/A	N/A	N/A
FAST-LIO2	3.974	3.485	8.558	0.028	0.007	2.756	0.003	0.046	0.133
ORB-SLAM3 Stereo	2.587	2.421	4.663	0.211	0.142	N/A	N/A	N/A	N/A
ORB-SLAM3 S+IMU	163.928	145.218	344.464	N/A	N/A	N/A	N/A	N/A	N/A
cuVSLAM	0.712	0.656	1.345	0.007	0.006	3.293	0.003	0.024	0.061
lidarslam_ros2	0.034	0.032	0.058	0.007	0.006	0.327	0.018	0.234	0.512
EKF	0.007	0.003	0.051	0.009	0.003	0.085	0.195	0.723	0.978

4.4.1 Trajectory Accuracy

4.4.2 Map Quality

4.4.3 Computational Cost

4.5 Discussion

4.5.1 LiDAR-Based Methods

4.5.2 LiDAR-IMU Methods

4.5.3 Visual Methods

4.5.4 Failure Cases

Chapter 5

Conclusions

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Appendix A

Appendices